DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY
University Examinations 2018/2019
SECOND YEAR **SEMESTER** II EXAMINATIONFOR THE DEGREE OF **BACHELOR OF SCIENCEIN INFORMATION TECHNOLOGY**

CIT 2205: Mobile Computing

DATE: AUGUST 2020 **TIME: 2 HOURS**

Instructions: Answer Question 1 and Any Other Two.

Question 1: (30 Marks)

a) In reference to wireless communication, explain a pervasive computing system (2mks)

b) Describe how a wireless signal is propagated in a LAN communication (2mks)

c) Explain that basic operating principle of a modem during wireless transmission (2mks)

d) Highlight TWO issues that influence quality signalling, in the process of a sharing a wireless communication channel (2mks)

e) Highlight the key conditions, which might trigger a handoff management during a phone call in a mountainous region (2mks)

f) Highlight the main limitation of an ad-hoc mobile network (2mks)

g) Explain TWO reasons why it is not advisable to use plugins, in a mobile application's micro- browsers (2mks)

h) Highlight a key function of Sub-network Dependent Convergence Protocol, in transmission of IP data packets. (2mks)

i) In reference to a video conference call, state a key feature of UDP protocol that is not supported by TCP protocol? (2mks)

j) In network programming, explain the use of a network interface (3mks)

k) Define the term pulse-code modulation as used in GSM communication (2mks)

l) List TWO supplementary services available to standard mobile communication features (2mks)

m) Describe the basic use WML in a WAP environment (2mks)

n) Describe what is meant by a 'tunnel' in a virtual private networking application (3mks)

QUESTION TWO (20 MARKS)

a) Describe how a sketch architectural design of wireless sensor communication platform can be implemented to detect smoke on a crops farm (4mks)
b) Describe how the WAP topology supports use of online integration of internet enabled applications (4mks)
c) In reference to any TWO computing paradigms, discuss why it would be convenient to develop an e-learning mobile application, for high school educational activities (4mks)
d) Using a sketch, describe what how satellite communication, is applied by a local digital T.V broadcasting organization (4mks)
e) In reference to an GSM communication process, explain any TWO subsystems that provide international roaming (4mks)

QUESTION THREE (20 MARKS)

a) In reference to the term 'mobility management', discuss TWO techniques that a Kenyan Network Service provider would use to guarantee reliable mobile network signals (6mks)
b) With reasons, explain any TWO protocols that facilitate wired communication in a corporate cellular digital packets data (CDPD) network (4mks)
c) Discuss THREE kinds of servers that enable a message to be transmitted via a Skype video conference system architecture (6mks)
d) In reference to mobile network security, describe TWO security issues that a mobile application developer should consider implementing a covid-19 tracking application (4mks)

QUESTION FOUR (20 MARKS)

a) Describe the working of Mobile IP communication components, say between Kenya and Nigerian sub-network regions (6mks)
b) In reference to Mobile Network Architecture, describe how a digital mobile bank wallet functions, while ensuring synchrony with an ATM system of a local bank (4mks)
c) Using a diagram, explain how mobile phone signal propagation applies multi-path signal propagation, to statistically overcome interference (6mks)
d) In reference to satellite internet communication service, sketch and explain TWO unique components, that support data transmission, in a Google internet balloon (4mks)

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